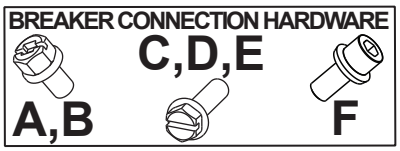
SIEMENS
P1-250 Torque Instructions

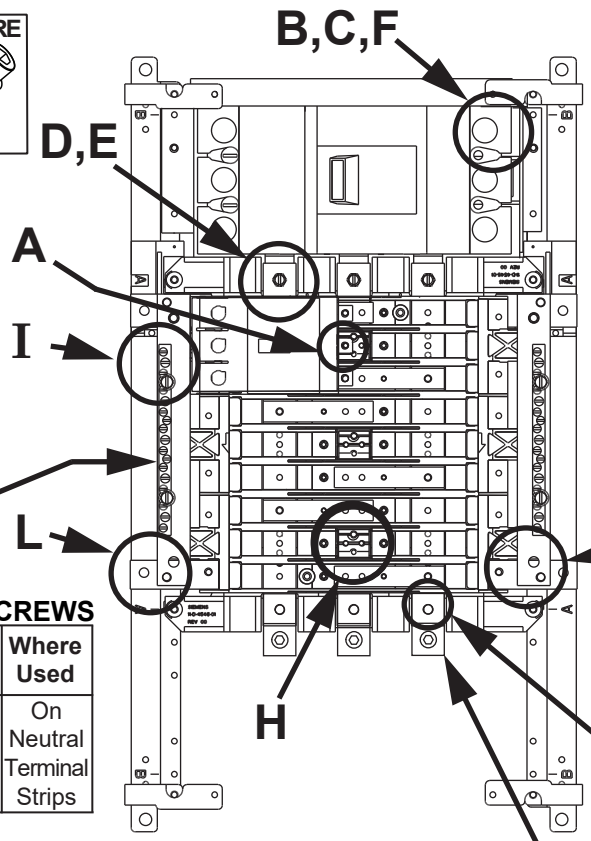
P1-400 version on back of page

This equipment has been tested & inspected before shipment. However, shipping conditions may have loosened some connections. **Before energizing**, it is **strongly recommended** that the tightness of all accessible connections be checked. When values are not marked on the devices, torque all connectors to the values indicated below.

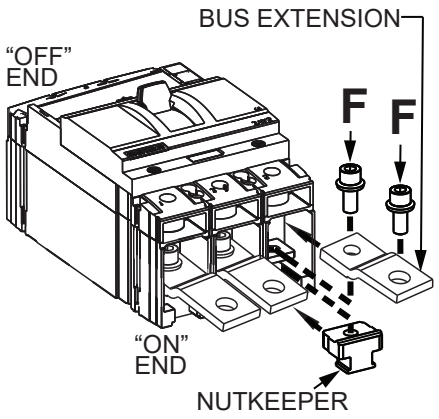
BREAKER CONNECTIONS

Thread Size	Torque (lb-in)	Driver Size & Type	Where Used	Item
#10-32	20	#2 Square Drive, 5/16" Nut Driver or Standard Flat-tip	Branch breaker to phase bus	A
			BL / BT / BQD / ED / xGB / 3VA41 breaker to straps	B
1/4"-20	72	7/16" Nut Driver or Standard Flat-tip	FD / QJ / QR breaker to straps	C
			BL / BT / BQD / ED / FD / xGB / QJ / QR / 3VA41 / 3VA52 / 3VA61 / 3VA62 straps to phase bus	D
			Main copper lugs to phase bus	E
5/16"-18	177	1/4" Allen Wrench or Hex Key Drive	3VA52 / 3VA61 / 3VA62 breaker to bus extensions or bus extensions to straps	F



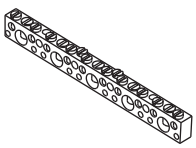


(isometric view of 3VA breaker showing BUS EXTENSIONS and NUTKEEPERS)



BREAKER MOUNTING SCREWS

Breaker Type	Slotted-head Screw	Torque (lb-in)
QR	1/4"-20 x 3-1/4"	35
FD	1/4"-20 x 4"	40
3VA52 3VA61 3VA62	#8-32 x 3"	13



BRANCH NEUTRAL WIRE SCREWS

Wire Size (AWG)	Torque (lb-in) SMALL Hole	Torque (lb-in) LARGE Hole	Driver Size & Type	Where Used
#14 - #10	20	35	Standard Flat-tip / #2 Square Drive	On Neutral Terminal Strips
#8	25	40		
#6	35	45		
#4	-	45		
#3 - 2/0	-	50		

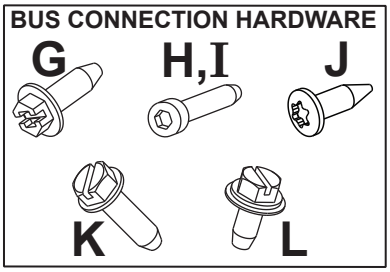
3VA NUTKEEPERS

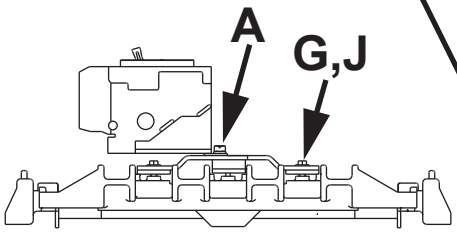
Breaker Type	Torque (lb-in)
3VA52 3VA61 3VA62	177



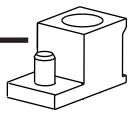
BOLT / NUT SETS

Thread Size	Torque (lb-in)	Driver Size & Type	Where Used
1/4"-20	72	7/16" Nut Driver	To Secure Lugs to Bus





(bottom end view minus some components for clarity)



LUG SET SCREWS

Allen Wrench Size	Torque (lb-in)	Driver Type
5/16"	275	Allen Wrench or Hex Key
3/8"	375	
1/2"	500	

BUS CONNECTIONS

Thread Size	Torque (lb-in) per Bus Material* 1/8"CU	3/16"AL	Driver Size & Type	Where Used	Item
#10-24	25	30	#2 Square Drive, 1/4" Nut Driver or Std. Flat-tip	A & C-phase straps to phase bus	G
			5/32" Hex Key Drive or Allen Wrench	B-phase connectors to phase bus	H
				Neutral terminal strips to risers	I
	30	30	T25 TORX Drive	A & C-phase straps to phase bus	J
1/4"-20	50	30	3/8" Nut Driver or Std. Flat-tip	2/0 Neutral strips to risers, Risers to crossbus/lug pads	K
5/16"-18	72	108	1/2" Nut Driver	Copper or compression neutral lugs to crossbus/lug pads	L

*** Bus Material type can be determined by: 1. Measuring thickness of phase bus at exposed ends with calipers OR**

2. Review catalog number on rating label and note the last characters:

a. If it ends in "...A" or "...AT", the "A" stands for Aluminum

b. If it ends in "...C" or "...CT", the "C" stands for Copper

SIEMENS
P1-400 Torque Instructions

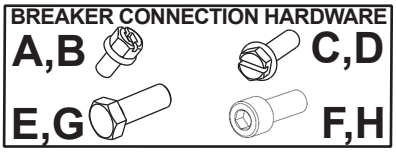
P1-250 version on back of page

This equipment has been tested & inspected before shipment. However, shipping conditions may have loosened some connections. **Before energizing**, it is **strongly recommended** that the tightness of all accessible connections be checked. When values are not marked on the devices, torque all connectors to the values indicated below.

BREAKER CONNECTIONS

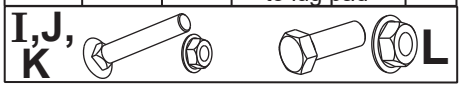
Thread Size	Torque (lb-in)	Driver Size & Type	Where Used	Item
#10-32	20	#2 Square Drive, 5/16" Nut Driver or Standard Flat-tip	Branch breaker to phase bus	A
			BL / BT / BQD / ED / xGB / 3VA41 breaker to straps	B*
1/4"-20	72	7/16" Nut Driver or Standard Flat-tip	FD / QJ / QR breaker to straps	C*
			BL / BT / BQD / ED / FD / xGB / QJ / QR / 3VA41 / 3VA52 / 3VA61 / 3VA62 straps to phase bus	D*
5/16"-18	144	1/2" Nut Driver	Main lugs to phase bus	E
	177	1/4" Allen Wrench or Hex Key Drive	3VA52 / 3VA61 / 3VA62 breaker to bus extensions or bus extensions to straps	F
3/8"-16	240	9/16" Nut Driver	JD breaker to straps	G
	354	5/16" Allen Wrench or Hex Key Drive	3VA53 / 3VA63 breaker to bus extensions	H

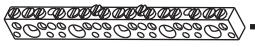
BREAKER CONNECTION HARDWARE



BOLT / NUT SETS

Thread Size	Torque (lb-in)	Nut Driver Size	Where Used	Item
1/4"-20	72	7/16"	Compression / copper lug pad to neutral riser	I
			Copper lug to lug pad	J
			Std. main neutral to neutral riser	K
1/2"-13	500	11/16"	Compression lug to lug pad	L

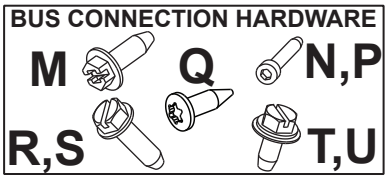


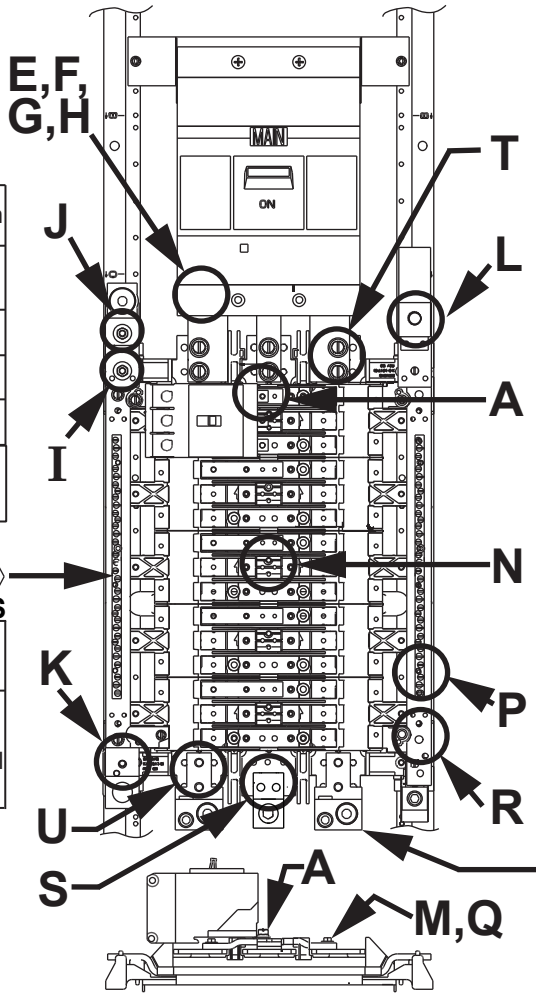


BRANCH NEUTRAL WIRE SCREWS

Wire Size (AWG)	Torque (lb-in) SMALL Hole	Torque (lb-in) LARGE Hole	Driver Size & Type	Where Used
#14 -#10	20	35	Standard Flat-tip / #2 Square Drive	On Neutral Terminal Strips
#8	25	40		
#6	35	45		
#4	-	45		
#3 - 2/0	-	50		

BUS CONNECTION HARDWARE





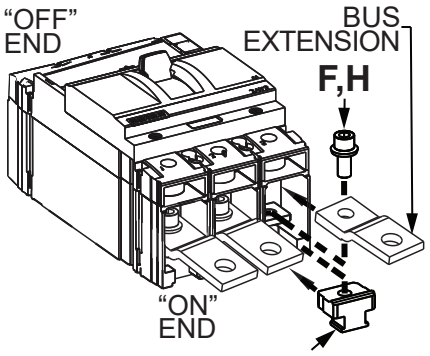
(bottom view minus some components for clarity)

BREAKER MOUNTING SCREWS

Breaker Type	Slotted-head Screw	Torque (lb-in)
QR	1/4"-20 x 3-1/4"	35
FD	1/4"-20 x 4"	40
3VA52 / 61 / 62	#8-32 x 3"	13

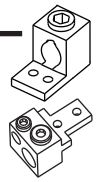
3VA NUTKEEPERS

Breaker Type	Torque (lb-in)
3VA52 / 3VA61 / 3VA62	177
3VA53 / 3VA63	354



(isometric view of 3VA breaker showing BUS EXTENSIONS and NUTKEEPERS)

LUG SET SCREWS



Allen Wrench Size	Torque (lb-in)	Driver Type
5/16"	275	Allen Wrench or Hex Key
3/8"	375	
1/2"	500	
9/16"	600	

BUS CONNECTIONS

Thread Size	Torque (lb-in) per Bus Material** 1/8"CU	3/16"AL	3/16"CU	1/4"AL	Driver Size & Type	Where Used	Item
#10-24	25	30	50	30	#2 Square Drive, 1/4" Nut Driver or Flat-tip	A & C-phase straps to phase bus	M
					5/32" Hex Key Drive or Allen Wrench	B-phase connectors to phase bus	N
						Neutral terminal strips to risers	P
	30	30	30	30	T25 Torx Drive	A & C-phase straps to phase bus	Q
1/4"-20	50	30	72	50	3/8" Nut Driver or Std. Flat-tip	Neutral strips to risers, Risers to crossbus/riser supports	R
						750kcmil main lugs to panel bus	S
5/16"-18	72	108	144	108	1/2" Nut Driver	JD / 3VA53 / 3VA63 straps to panel bus	T
						600kcmil main lugs to panel bus	U

*** The BL / BT / BQD / ED / FD / xGB / QJ / QR / 3VA41 / 3VA52 / 3VA61 / 3VA62 breakers all attach using the same hardware in the P1-400 as the P1-250. Therefore, refer to B, C & D image arrows on the other side of this page for those.**

**** Bus Material type can be determined by:**
1. **Measuring thickness of phase bus at exposed ends with calipers AND**
2. **Review catalog number on rating label and note the last characters:**
   a. **If it ends in "...A" or "...AT", the "A" stands for Aluminum**
   b. **If it ends in "...C" or "...CT", the "C" stands for Copper**